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SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

INTRODUCTION TO RADAR SYSTEMS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Draw the block diagram of Radar System and explain the function of each block in detail. **06**

b. i. In a Radar System, it is observed that the echo received is after 9.15μsec. Calculate the distance of the stationary object from the Radar. **4+4**

ii. Calculate the maximum range of radar which operates at a frequency of 10GHz, peak pulse power of 600kW. If the antenna effective area is $5m^2$, and the area of target is $20m^2$, minimum receiver power is 10^{-13} Watt.

2. a. Why there is a requirement of Isolation between Transmitter and Receiver in CW Radar System? What are the different approaches to provide isolation? **08**

b. What is importance of Non-zero IF Receiver in CW Radar System? Write a short note on FM CW Radar. **06**

3. a. Draw the block diagram of MTR Radar with power amplifier, and explain the function of each block in detail. **06**

b. Write short notes on the following topic: **3+5**

i. Blind Speeds ii. Double Cancellation

OR

4. a. Draw the block diagram of MTR Radar using range gates and filters and explain the function of each block in detail. **08**

b. Write short notes on the following topic: **06**

i. Non - Coherent MTI's.

ii. Staggered PRFs.

5. a. Describe in detail about the concept of Sequential Lobing. **06**

b. Draw the block diagram of Monopulse Tracking Radar and explain the function of each block in detail. **08**

OR

6.	**a.**	Write short notes on the following topic:	**06**
		i. The Magnetron Oscillator.	
		ii. Klystron Amplifier.	
	b.	Draw the block diagram of Travelling Wave Tube and explain the function of each block in detail.	**08**
7.	**a.**	What are Radar Receivers? Explain its characteristics.	**06**
	b.	Explain in detail various types of Duplexers.	**08**
		OR	
8.	**a.**	Explain Phased Array Antennas with it's advantages and limitations.	**06**
	b.	Write short notes on the following topic:	**4+4**
		i. Noise Figure	
		ii. Noise Temperature	